Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 9: Trees, Cutsets and Fundamental Loops

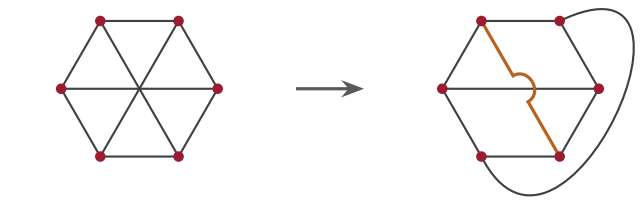
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Colour code used in these notes and in the slides

current i | voltage v, cutsets, loops | tree branches | nodes ①

1. Beyond planar graphs

Meshes need a drawing with no crossings. A nonplanar graph has no such drawing, so it has no meshes and no mesh matrix. In the hexagon below, the three diagonals cross at the centre. Moving one diagonal outside removes one crossing, but a jumper wire (a hop) is still needed.



crossings at the centre jumper wire: still one hop

How do we then find $n-1$ independent KCL and $b-n+1$ independent KVL equations? This lecture gives a method that needs no planar drawing and works for **any connected graph**.

2. Trees

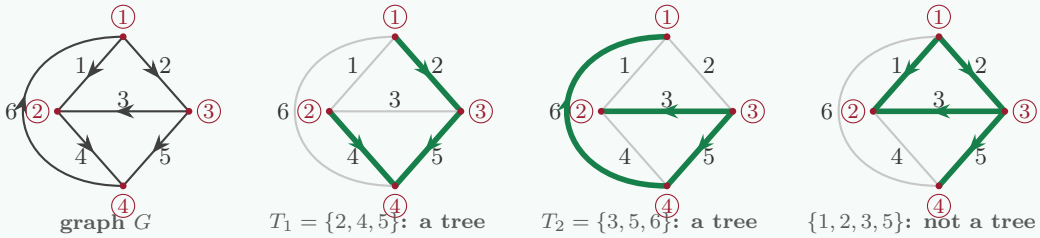
Tree

Let G be a connected graph. A subgraph T of G is a **tree** if

1. T contains all the nodes of G,
2. T is connected, and
3. T contains no loops.

Example 9.1 — trees of the bridge graph

Which of the green subgraphs are trees?



T_1 and T_2 reach all four nodes, are connected and have no loop. The set $\{1,2,3,5\}$ contains the loop of branches 1, 2, 3, so it is not a tree. A graph has many trees.

A tree has $n-1$ branches

Build the tree one branch at a time, starting from a single node. Each new branch must reach a new node: a branch between two nodes already reached would close a loop. All n nodes must be reached, so the tree has exactly $n-1$ branches. The other $b-n+1$ branches are left out.

3. Cutsets

Cutset

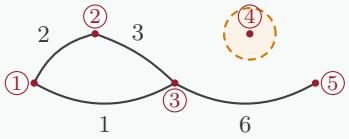
Let G be a connected graph. A set C of branches is a **cutset** if

1. removing all the branches of C leaves the graph unconnected, and
2. removing all the branches of C except any one of them leaves it connected.

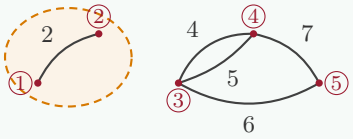
The second condition makes the cutset minimal: every branch in it is needed. Minimal does not mean fewest branches; cutsets of the same graph can have different sizes.

Example 9.2 — cutsets of a figure-eight graph

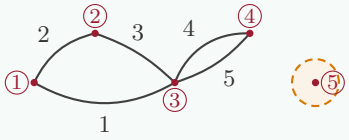
In each panel the removed branches are deleted; look at what is left.



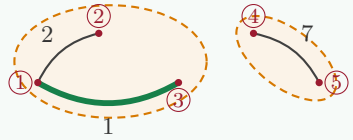
$C_1 = \{4, 5, 7\}$: node ④ cut off



$C_2 = \{1, 3\}$: nodes ①, ② cut off



$C_3 = \{6, 7\}$: node ⑤ cut off



$\{1, 3, 4, 5, 6\}$ with 1 put back: still two pieces

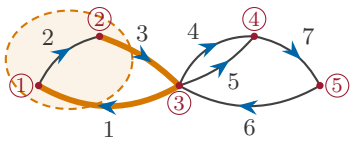
For C_1: removing branches 4, 5, 7 cuts off node ④, so condition 1 holds. Putting back any one of them reaches node ④ again, so condition 2 holds. C_2 and C_3 are checked the same way. With all branches back, the dashed closed surface around the cut-off piece crosses exactly the branches of the cutset.

The set $\{1, 3, 4, 5, 6\}$ is *not* a cutset. Removing it leaves three pieces. With branch 1 put back, the graph is still unconnected, so condition 2 fails. Branches 1 and 3 are not needed: $\{4, 5, 6\}$ alone is a cutset.

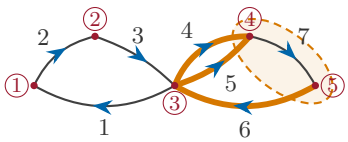
The set $\{4, 6\}$ is not a cutset either, but for the opposite reason. Removing it leaves the graph connected, since branch 5 still joins nodes ③ and ④, so condition 1 fails. Branches 4 and 5 are parallel: both join ③ and ④. A closed surface that crosses one of them crosses the other too. So parallel branches are in a cutset together or not at all.

KCL on a cutset. A cutset splits the graph into two parts. A closed surface around one part crosses exactly the branches of the cutset. So by KCL on that closed surface, the currents of the cutset branches add up to zero (each with + if it leaves the surface, − if it enters).

Take the branch currents below, in the arrow directions. They satisfy KCL at every node.



$C_2 = \{1, 3\}$



$\{4, 5, 6\}$

branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

For each cutset, write leaving currents with $+$ and entering ones with $-$:

$$\begin{aligned} C_1 = \{4,5,7\}: &\quad i_7 - i_4 - i_5 = 3 - 1 - 2 = 0, \\ C_2 = \{1,3\}: &\quad i_3 - i_1 = 2 - 2 = 0, \\ \{4,5,6\}: &\quad i_6 - i_4 - i_5 = 3 - 1 - 2 = 0. \end{aligned}$$

For C_1 this is just KCL at node ④. For C_2 it is the sum of KCL at nodes ① and ②: $(i_2 - i_1) + (i_3 - i_2) = 0$. The current i_2 is inside the surface and cancels.

4. Fundamental cutsets

Fundamental cutset and cotree

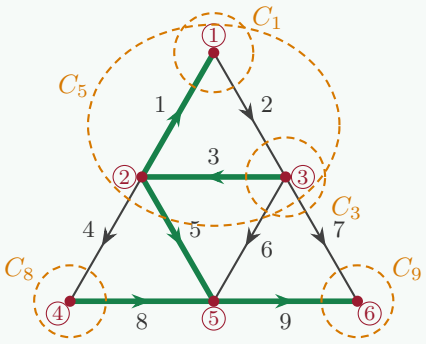
Let T be a tree of a connected graph G, and β a branch of T. There is exactly one cutset of G that contains β and no other tree branch. It is the **fundamental cutset** of β.

The branches of G that are not in T form the **cotree**.

Removing the tree branch β splits the tree into two parts. The fundamental cutset is the set of branches that join these two parts: β itself and some cotree branches. Its direction is taken as the direction of β.

Example 9.3 — fundamental cutsets and KCL

The graph G has $n = 6$ nodes and $b = 9$ branches. Take the tree $\{1,3,5,8,9\}$ (green, $n-1 = 5$ branches). The cotree is $\{2,4,6,7\}$ ($b - n + 1 = 4$ branches).



tree br.	cutset	KCL
1	$\{1,2\}$	$i_1 - i_2 = 0$
3	$\{3,2,6,7\}$	$i_3 - i_2 + i_6 + i_7 = 0$
5	$\{5,4,6,7\}$	$i_5 + i_4 + i_6 + i_7 = 0$
8	$\{8,4\}$	$i_8 - i_4 = 0$
9	$\{9,7\}$	$i_9 + i_7 = 0$

Each surface encloses one of the two parts left after removing a tree branch. For C_5: removing branch 5 leaves the parts $\{①,②,③\}$ and $\{④,⑤,⑥\}$. Branches 5, 4, 6, 7 join them, and all four go from the upper part to the lower part, the same way as branch 5: every term has $+$.

KCL on the fundamental cutsets

A tree gives $n-1$ fundamental cutsets, so $n-1$ KCL equations. They are independent: each equation contains its own tree branch current, which appears in no other equation.

Remark. Each equation has one tree branch current. So once the cotree branch currents are known, the tree branch currents follow. For G:

$$i_1 = i_2, \quad i_3 = i_2 - i_6 - i_7, \quad i_5 = -i_4 - i_6 - i_7, \quad i_8 = i_4, \quad i_9 = -i_7.$$

5. Fundamental loops

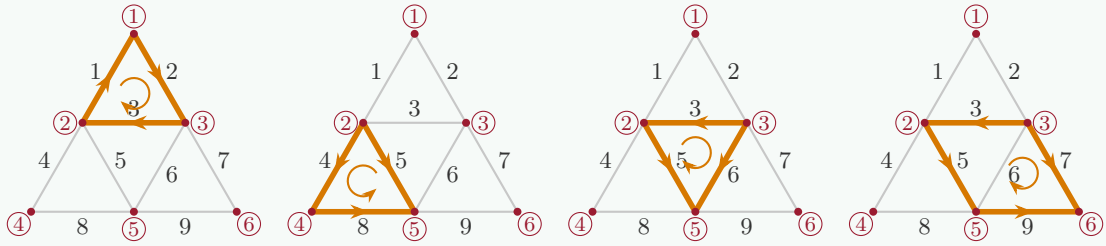
Fundamental loop

Let T be a tree of a connected graph G. Each cotree branch defines exactly one loop that contains that cotree branch and no other cotree branch. It is the **fundamental loop** of that cotree branch. Its direction is taken as the direction of the cotree branch.

Adding one cotree branch to the tree closes exactly one loop: the cotree branch plus the tree path between its two ends.

Example 9.4 — fundamental loops and KVL

Same graph and tree as in Example 9.3. Each cotree branch closes one loop. Walk it in the direction of its cotree branch: a branch voltage counts + if its arrow points along the walk, − if against.



loop of 2: $\{2,3,1\}$ $v_2 + v_3 + v_1 = 0$

loop of 4: $\{4,8,5\}$ $v_4 + v_8 - v_5 = 0$

loop of 6: $\{6,5,3\}$ $v_6 - v_5 - v_3 = 0$

loop of 7: $\{7,9,5,3\}$ $v_7 - v_9 - v_5 - v_3 = 0$

KVL on the fundamental loops

A tree gives $b-n+1$ fundamental loops, so $b-n+1$ KVL equations. They are independent: each equation contains its own cotree branch voltage, which appears in no other equation.

Remark. Each equation has one cotree branch voltage. So once the tree branch voltages are known, the cotree branch voltages follow. For G:

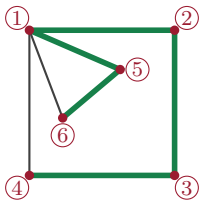
$$v_2 = -v_1 - v_3, \quad v_4 = v_5 - v_8, \quad v_6 = v_3 + v_5, \quad v_7 = v_3 + v_5 + v_9.$$

6. Any connected graph

The tree is the key, and every connected graph has one. So:

	how many	gives
tree branches	$n-1$	fundamental cutsets: the $n-1$ KCL equations
cotree branches	$b-n+1$	fundamental loops: the $b-n+1$ KVL equations

No drawing is needed. This works for nonplanar graphs, and it is also the fix for separable graphs met earlier: the triangle inside a square, attached at one node. Pick a tree, for example the green one below. The two cotree branches give two fundamental loops, the square and the triangle, and each is a single loop.



7. Looking ahead

Node voltages and mesh currents were good unknowns: few, and one law held automatically. Trees give the general version. The $n-1$ **tree branch voltages** fix every branch voltage, like node voltages. The $b-n+1$ **cotree branch currents** fix every branch current, like mesh currents. Both work for any connected graph.

Summary

- A **tree** of a connected graph contains all nodes, is connected and has no loop. It has $n-1$ branches; the other $b-n+1$ form the **cotree**.
- A **cutset** is a minimal set of branches whose removal cuts the graph in two. KCL holds on every cutset: it is KCL on a closed surface.
- Each tree branch has a **fundamental cutset**: it and some cotree branches. These give $n-1$ independent KCL equations. Cotree currents fix the tree currents.
- Each cotree branch has a **fundamental loop**: it and a tree path. These give $b-n+1$ independent KVL equations. Tree voltages fix the cotree voltages.
- No drawing is needed: this works for any connected graph, planar or not, separable or not.

Next lecture

Duality: the dual of a planar graph and the dual of a circuit.

Reading

- Chua, Desoer & Kuh, *Linear and Nonlinear Circuits* — trees, cutsets and fundamental loops.
- Lecture notes and videos of Prof. Emre Tuna, `users.metu.edu.tr/etuna/ee201` — notes pp. 17–18; Lecture 8 video: nonplanar circuits 15:51, tree 16:10, example 18:01, cutset 22:33, example 25:32, fundamental cutset 30:35, example and cotree 32:55, KCL on fundamental cutsets 38:32, remark on cotree currents 41:48, fundamental loop 43:51, example 47:07, remark on tree voltages 55:38.